

Time Series Analysis 2026

CAPSTONE PROJECT

Data-Driven Stock Analysis using Time Series Models on StockGro

Virtual Capital: ₹10,00,000 | Platform: StockGro (NSE Live Data)

Trading Window

14 May 2026 (Thursday) → 15 May 2026 (Friday)

Net Portfolio Return: +1.20% | Net P&L: +₹12,028.86

Report Prepared By –

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1. Methodology and Models Used

This project applies an end-to-end time series analysis pipeline to NSE-listed stocks, progressing from data collection and preprocessing through forecasting, volatility analysis, portfolio construction, live virtual trading execution on StockGro, and a final comparison of predicted versus actual outcomes.

ARIMA (Auto-Regressive Integrated Moving Average) was selected as the primary forecasting model due to its interpretability, computational efficiency, and proven suitability for classical financial time series. Model parameters were tuned via AIC/BIC grid search over candidate (p, d, q) combinations.

Pipeline Summary

- Data: Yahoo Finance via yfinance, January 2021 to December 2025, daily interval (1d)
- Stationarity: ADF test applied to all series; first-order differencing where p-value ≥ 0.05
- Split: Training set up to June 2025; test set July to December 2025 (minimum 6 months)
- Modelling: ARIMA (5,1,2) fitted on training set; evaluated with RMSE, MAPE, Directional Accuracy
- Forecasting: 2-day ahead point forecasts with 95% confidence intervals using full 2021-2025 data
- Portfolio: 5 NSE stocks across 5 sectors; Forecast-Guided and Volatility-Aware strategies
- Execution: Trades placed on StockGro on Day 1 (14 May); portfolio recorded at Day 2 close (15 May)
- Evaluation: ARIMA forecasts compared against actual StockGro closing prices; P&L and directional accuracy assessed.

2. Stock Selection Rationale

A rolling standard deviation screening was conducted on 8 NSE candidate stocks before finalising the portfolio universe. The average 30-day rolling standard deviation of log returns was computed and plotted for each candidate. Five stocks were selected to maximise sector diversity while covering a spectrum of volatility profiles.

Selected Stock Universe

Stock Name	Stock	Ticker	Sector	Volatility	Selection Rationale
HDFC Bank	HDFCBANK	HDFCBANK.NS	Banking	Low	Lowest rolling volatility; sector stability; anchor position
Tata Consultancy Services	TCS	TCS.NS	IT	Moderate	Reliable IT bellwether; consistent long-term uptrend
Hindustan Unilever	HINDUNILVR	HINDUNILVR.NS	FMCG	Low	Defensive sector; low correlation to cyclicals
Sun Pharmaceuticals	SUNPHARMA	SUNPHARMA.NS	Pharma	Moderate	Sector diversification; pharma recovery momentum
Mahindra & Mahindra	M&M	M&M.NS	Automobiles	Higher	Growth sector exposure; auto momentum 2023-2025

This selection spans five uncorrelated sectors, reducing portfolio-level concentration risk. HDFCBANK and HINDUNILVR serve as low-volatility defensive anchors while M&M provides growth exposure with a correspondingly smaller allocation.

3. Data Preprocessing Steps

3.1 Missing Value Handling

Data downloaded via yfinance is aligned to the NSE trading calendar. Forward fill was applied as a precaution to handle any gaps from trading halts or public holidays.

3.2 Stationarity Testing - Augmented Dickey-Fuller (ADF) Test

The ADF test was applied to raw closing price series for each stock. Decision rule: $p\text{-value} < 0.05$ implies stationarity; $p\text{-value} \geq 0.05$ requires differencing.

Stock	ADF Statistic	p-value	Initial Result	Post-Diff
TCS	-2.277568	0.179305	Non-Stationary	Stationary
HDFCBANK	-1.395872	0.584254	Non-Stationary	Stationary
HINDUNILVR	-2.881449	0.047548	Stationary	Stationary
SUNPHARMA	-0.847624	0.804731	Non-Stationary	Stationary
M&M	0.683759	0.989508	Non-Stationary	Stationary

All five stocks were non-stationary in raw price levels, as expected for financial data. First-order differencing ($d=1$) rendered all series stationary, confirming the integration order for ARIMA.

3.3 Normalisation

Normalisation was not required as ARIMA operates on the raw differenced price series. Scaling would have been applied had LSTM been implemented.

3.4 Train-Test Split

- Training Set: January 2021 to June 2025 (approximately 4.5 years of daily data)
- Test Set: July 2025 to December 2025 (6 months, satisfying the minimum project requirement)

4. Forecast Results with Confidence Intervals

ARIMA (5,1,2) was trained on the full dataset (January 2021 to December 2025) and used to generate 2-day ahead price forecasts for the live trading window. The `get forecast()` method provided both point estimates and 95% confidence intervals.

Test Set Evaluation (July to December 2025)

Stock	RMSE	MAPE	Dir. Accuracy	Comment
HDFCBANK	19.5245	0.0172	0.5026	Most reliable forecast
TCS	336.3127	0.1048	0.5161	Stable performance
HINDUNILVR	217.4573	0.0762	0.4991	Consistent
SUNPHARMA	68.3366	0.0328	0.5000	Acceptable
M&M	388.1831	0.0934	0.4950	Highest uncertainty

2-Day Ahead Price Forecast

Stock	Day 1 Pred (₹)	Day 2 Pred (₹)	Direction	Day 1 CI Width	Day 2 CI Width	95% CI Width	Confidence
TCS	3189.60	3189.69	Up ↑	168.68	242.16	Wide	Stable
HDFCBANK	991.08	990.68	Down ↓	39.72	56.92	Narrow	Most stable
HINDUNILVR	2290.70	2292.65	Up ↑	114.14	161.30	Wide	Moderate
SUNPHARMA	1709.97	1709.09	Down ↓	60.29	82.51	Narrow	Moderate
M&M	3657.91	3658.52	Up ↑	138.01	193.68	Wide	Highest uncertainty

ARIMA predicted marginal downward movement for all stocks over the 2-day window. This is characteristic of ARIMA's mean-reverting behaviour in ultra-short forecast horizons. Confidence intervals are widest for M&M, consistent with its higher rolling volatility.

5. Volatility and Trend Analysis Findings

5.1 Log Return Volatility

Log returns were computed as $\ln(P_t / P_{t-1})$. A 20-day rolling standard deviation captured time-varying short-term volatility. These values directly informed the Volatility-Aware allocation weights in Task 5.

Stock	Avg Rolling Vol	Risk Profile	Portfolio Implication
HDFCBANK	Lowest	Low	Largest allocation (29%)
HINDUNILVR	Low	Low	Second defensive position (22%)
TCS	Moderate	Medium	Core IT holding (25%)
SUNPHARMA	Moderate	Medium	Small allocation (5%)
M&M	Highest	High	Reduced allocation (19%)

5.2 Correlation Analysis

A correlation matrix of log returns was computed across all five stocks. No pair exceeded a correlation of 0.7, confirming low-to-moderate inter-stock relationships. HDFCBANK and HINDUNILVR showed the lowest correlations with the rest, reinforcing their role as defensive diversifiers and validating the portfolio construction rationale.

5.3 STL Decomposition for All 5 Stocks

Seasonal and Trend decomposition using Loess (STL) was applied to all five stocks with a 30-day period, separating each series into trend, seasonality, and residual components.

Stock	Trend	Observation	Implication
HDFCBANK	Upward	Consistent long-term appreciation 2021-2025	Core holding confirmed
TCS	Upward	Strong sustained uptrend across full period	Reliable growth trajectory
HINDUNILVR	Sideways/Up	Mild upward bias, low seasonal strength	Stability overgrowth
SUNPHARMA	Upward	Recovered strongly post-2022 correction	Positive momentum
M&M	Upward	Sharp uptrend 2022-2025 with high residual noise	Growth but volatile

6. Portfolio Composition and Allocation Rationale

Portfolio construction applied two strategies: Strategy A (Forecast-Guided Allocation) and Strategy B (Volatility-Aware Sizing). Total virtual capital of ₹10,00,000 was deployed with ₹9,97,633.51 invested.

Strategy A - Forecast-Guided Allocation

Stocks with lower MAPE on the test set and tighter confidence intervals received larger allocations. HDFCBANK and TCS, which demonstrated the most stable forecasting behaviour, received the two largest weights.

Strategy B - Volatility-Aware Sizing

Weights were adjusted inversely proportional to predicted rolling volatility using $w_i = (1/\sigma_i) / \sum_j (1/\sigma_j)$. M&M, with the highest volatility, received a reduced weight (19%) despite its upward trend. SUNPHARMA received the smallest allocation at 5%.

Stock	Weight	Invested (₹)	Qty	Rationale
HDFCBANK	29%	2,88,750.81	389	Lowest volatility anchor; highest forecast confidence; Banking sector
TCS	25%	2,52,236.40	115	Core IT holding; moderate volatility; strong consistent trend
HINDUNILVR	22%	2,24,361.00	100	Defensive FMCG; low correlation to cyclical; stability focus
M&M	19%	1,84,899.00	60	Auto growth exposure; weight reduced for higher volatility
SUNPHARMA	5%	47,386.30	26	Sector diversification; smallest allocation; moderate volatility
TOTAL	100%	9,97,633.51	690	

7. Model Comparison and Accuracy Evaluation

Three forecasting approaches were assessed. ARIMA was fully implemented and used for all portfolio decisions. LSTM and Prophet were evaluated conceptually.

Model	Type	Strengths	Weaknesses	Used?
ARIMA (5,1,2)	Statistical	Interpretable; AIC/BIC tunable; fast; suited to stationary series	Assumes linearity; mean-reverts in short horizons; no macro inputs	Yes
LSTM / GRU	Deep Learning	Captures nonlinear patterns and long-range temporal dependencies	Requires large data; heavy tuning; computationally expensive	No
Prophet	Additive	Handles seasonality automatically; robust to missing data	Can oversimplify volatile financial series	No

Why ARIMA Was Selected

- AIC/BIC grid search over $p \in \{1,2,3,5\}$, $d=1$, $q \in \{0,1,2\}$ identified ARIMA (5,1,2) as optimal for TCS training data
- Same order applied to all five stocks for consistency and fair cross-stock comparison
- Residual diagnostics confirmed acceptable fit on training data
- Computationally efficient: all five models trained within the project timeline

Ensemble Consideration

A simple ARIMA + Prophet ensemble was considered but not implemented, as the 2-day forecast horizon is too short for meaningful ensemble gains. For a 5-10 day horizon, an ensemble would reduce systematic directional bias.

8. StockGro Execution Summary

All trades were executed on the StockGro virtual trading platform under the event portfolio “Portfolio – Time Series Analysis 2026.” The chosen execution window was Day 1: Thursday 14 May 2026 and Day 2: Friday 15 May 2026. All positions were opened before 3:25 PM IST on Day 1 and recorded at market close on Day 2.

Stock	Qty	Avg Buy (₹)	Purchase Val (₹)	Weight	Day 2 Close (₹)	P&L (₹)
SUNPHARMA	26	1,822.55	47,386.30	4.75%	1,848.08	+663.78
HINDUNILVR	100	2,243.61	2,24,361.00	22.5%	2,255.10	+1,149.00
M&M	60	3,081.65	1,84,899.00	18.5%	3,087.54	+353.40
TCS	115	2,193.36	2,52,236.40	25.3%	2,220.01	+3,064.75
HDFCBANK	389	742.29	2,88,750.81	28.9%	762.33	+7,795.56
TOTAL	690	–	9,97,633.51	100%	–	+13,026.49

- Gross unrealised P&L at Day 2 close: +₹13,026.49
- Transaction charges deducted by StockGro (0.1%): –₹997.63
- Net realised profit: +₹12,028.86
- Net 2-day portfolio return: +1.20%

9. Predicted vs Actual Outcomes

This is the core evaluative section. ARIMA forecasts for Day 1 and Day 2 are compared against actual closing prices from StockGro, followed by directional accuracy analysis and portfolio P&L breakdown.

9.1 Price Level Comparison

The table below compares ARIMA model forecasts against actual NSE closing prices on StockGro. Directional accuracy is measured as: did the model correctly predict whether the Day 2 closing price was higher or lower than the Day 1 closing price?

Stock	D1 Predicted (₹)	D1 Actual (₹)	D1 Error %	D2 Predicted (₹)	D2 Actual (₹)	D2 Error %	Direction Correct?
HDFCBANK	991.08	759.28	30.53%	990.68	762.33	29.95%	No
TCS	3189.60	2,204.19	44.77%	3,189.69	2,220.01	43.68%	Yes ✓
HINDUNILVR	2290.70	2,227.14	2.85%	2292.65	2,255.10	1.67%	Yes ✓
SUNPHARMA	1709.97	1,858.28	7.99%	1709.09	1,848.08	7.52%	Yes ✓
M&M	3657.91	3,142.87	16.39%	3658.52	3,087.54	18.49%	No

Directional Accuracy: 3/5 stocks (60%)

- Correct: TCS (predicted Up → closed Up), HINDUNILVR (predicted Up → closed Up), SUNPHARMA (predicted Down → closed Down)
- Incorrect: HDFCBANK (predicted Down, market closed Up), M&M (predicted Up, market closed Down)

Note on price-level errors: Large percentage errors for HDFCBANK and TCS reflect a mismatch between historical adjusted prices used for model training versus actual NSE market prices in May 2026. This is a known data pipeline consideration when using yfinance across multi-year windows with corporate actions. HINDUNILVR showed the lowest price-level error (~1.67–2.85%), indicating the most accurate absolute forecast.

9.2 Portfolio Performance

The table below shows actual portfolio P&L measured from the average buy price to the Day 2 closing price. This is independent of the directional accuracy analysis above — the buy price and the Day 1 closing price are different values, so no direction comparison is included here.

Stock	Buy Price (₹)	Day 2 Close (₹)	Return %	Invested (₹)	Weight (%)	P&L (₹)
HDFCBANK	742.29	762.33	+2.70%	2,88,750.81	28.94%	+7,795.56
TCS	2,193.36	2,220.01	+1.22%	2,52,236.40	25.28%	+3,064.75
HINDUNILVR	2,243.61	2,255.10	+0.51%	2,24,361.00	22.49%	+1,149.00
M&M	3,081.65	3,087.54	+0.19%	1,84,899.00	18.53%	+353.40
SUNPHARMA	1,822.55	1,848.08	+1.40%	47,386.30	4.75%	+663.78
TOTAL	—	—	+1.31% gross	9,97,633.51	100%	+13,026.49

All five positions were profitable. HDFCBANK generated the highest absolute P&L (₹7,795.56), consistent with its role as the largest allocation and confirming the volatility-aware strategy. The average buy prices across all stocks were below the Day 2 closing prices, resulting in positive returns on every position.

Transaction charges of ₹997.63 (0.1%) were deducted by StockGro, bringing the net realised profit to ₹12,028.86, a net return of +1.20% over the 2-day window.

9.3 Summary Statistics

Metric	Value
Capital Allocated	₹10,00,000
Capital Deployed	₹9,97,633.51
Gross P&L (before charges)	+₹13,026.49
Transaction Charges (0.1%)	−₹997.63
Net P&L (after charges)	+₹12,028.86
Net Portfolio Return (2-day)	+1.20%
Stocks with Positive Returns	5 / 5
Model Directional Accuracy (Day1→Day2 close)	3 / 5 (60%) — TCS, HINDUNILVR, SUNPHARMA correct
Portfolio Positions Profitable	5 / 5 (all positions in profit)

Note: Model directional accuracy and portfolio profitability are two distinct measures. Directional accuracy evaluates the model in isolation — whether the predicted price movement direction (Day 1 forecast → Day 2 forecast) matched actual market movement (Day 1 close → Day 2 close). Portfolio profitability reflects the P&L from average buy price to Day 2 close. All 5 positions were profitable regardless of directional accuracy because the purchase prices were below Day 2 closing prices across all stocks.

10. Reflections

What Worked

- Portfolio diversification across five uncorrelated sectors was highly effective — all five positions delivered positive returns with no single stock dragging performance.
- Volatility-aware allocation correctly identified HDFCBANK as the safest anchor and assigned it the largest weight (29%), which also generated the highest absolute P&L (+₹7,795.56).
- The three lowest-volatility stocks (HDFCBANK, HINDUNILVR, TCS) accounted for 76% of capital and contributed ₹12,005.31 of the total gross P&L of ₹13,026.49.
- AIC/BIC grid search provided a principled, reproducible basis for ARIMA order selection rather than arbitrary parameter choice.
- STL decomposition across all five stocks confirmed upward trend trajectories that supported the bullish portfolio stance.
- Transaction cost accounting (₹997.63) ensured the reported net return accurately reflected real execution conditions.

What Did Not Work

- ARIMA's 2-day directional forecast was incorrect for all 5 stocks. The model predicted marginal downward movement, but the market moved up uniformly. ARIMA's mean-reverting behaviour in ultra-short horizons makes it poorly suited to anticipating broad market rallies.
- Predicted price levels showed large percentage errors for HDFCBANK and TCS, suggesting a possible data pipeline issue with adjusted vs unadjusted closing prices across the multi-year training window.
- Only one forecasting model was implemented. A single model provides no redundancy; its systematic biases (as seen here) have no corrective signal.

What Would Be Improved

- Implement Facebook Prophet as a second model and create a simple ensemble (average of ARIMA and Prophet) to reduce systematic directional bias.
- Incorporate exogenous variables such as Nifty 50 returns, sector index momentum, or macroeconomic indicators as additional ARIMAX regressors.
- Apply GARCH (1,1) for formal volatility forecasting to improve the precision of volatility-aware portfolio weights, especially for M&M.
- Use walk-forward (rolling) validation instead of a single static train-test split for more robust out-of-sample performance estimates.
- Extend the observation window to 5 trading days to reduce the impact of single-session noise on the model evaluation.

Key Takeaway

The project demonstrated that sound portfolio construction principles — sector diversification, volatility-aware sizing, and disciplined capital allocation — can deliver positive outcomes even when the underlying forecasting model shows directional limitations. The net return of +1.20% (₹12,028.86) over two trading days, with all five positions profitable, validates the portfolio strategy. The experience highlighted a critical insight: model accuracy and portfolio performance are related but distinct objectives, and a robust allocation framework can absorb forecast error while still generating returns.